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Stacked offset semiconductor package

Abstract

Disclosed is a semiconductor package comprising a first chip stack including on a substrate a plurality of first semiconductor chips in an offset stack structure and stacked to expose a connection region at a top surface of each of the first semiconductor chips, a second semiconductor chip on the substrate and horizontally spaced apart from the first chip stack, a spacer on the second semiconductor chip, and a second chip stack including third semiconductor chips in an offset stack structure on the first chip stack and the spacer. Each of the first semiconductor chips includes a first chip pad on the connection region and a first wire that extends between the first chip pad and the substrate. The first wire of an uppermost one of the first semiconductor chips is horizontally spaced apart from a lowermost one of the third semiconductor chips.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This U.S. nonprovisional application claims priority under 35 U.S.C § 119 to Korean Patent Application No. 10-2021-0095363 filed on Jul. 21, 2021 in the Korean Intellectual Property Office, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The present inventive concepts relate to a semiconductor package, and more particularly, to a semiconductor package including a stacked integrated circuit.

DISCUSSION OF RELATED ART

(3) A typical stacked package has a structure in which a number of substrates are stacked. For example, the stacked package may include semiconductor chips that are sequentially stacked on a printed circuit board (PCB). Connection pads are formed on the semiconductor chips. Bonding wires may be used to connect the connection pads, such that the semiconductor chips may be electrically coupled to each other. The printed circuit board is provided thereon with a logic chip that controls the semiconductor chips.

(4) Portable devices have been in increasing demand in recent electronic product markets, and as a result, reductions in size and weight of electronic parts mounted on the portable devices are desirable. In order to accomplish the reduction in size and weight of the electronic parts, there is a need for technology to integrate a number of individual devices into a single package as well as technology to reduce individual sizes of mounting parts. In particular, semiconductor packages operated at high frequency signals are required to have compactness and excellent electrical characteristics.

SUMMARY

(5) Some embodiments of present inventive concepts provide a compact-sized semiconductor package.

(6) Some embodiments of the present inventive concepts provide a semiconductor package with increased electrical properties.

(7) According to some embodiments of the present inventive concepts, a semiconductor package may include a first chip stack that includes a plurality of first semiconductor chips on a substrate, the plurality of first semiconductor chips being in an offset stack structure and being stacked to expose a connection region at a top surface of each of the first semiconductor chips, a second semiconductor chip on the substrate and horizontally spaced apart from the first chip stack, a spacer on the second semiconductor chip, and a second chip stack that includes a plurality of third semiconductor chips on the first chip stack and the spacer, the plurality of third semiconductor chips being in an offset stack structure. Each of the plurality of first semiconductor chips may include a first chip pad on the connection region, and a first wire that extends between the first chip pad and the substrate. The first wire of an uppermost one of the first semiconductor chips may be horizontally spaced apart from a lowermost one of the third semiconductor chips.

(8) According to some embodiments of the present inventive concepts, a semiconductor package may include a substrate, a first stack on the substrate, a second stack on the first stack, and a spacer on one side of the first stack and between the substrate and the second stack. The first stack may include a plurality of first semiconductor chips that are stacked on the substrate in ascending stepwise fashion offset in a first direction, a plurality of first wires that extend between the substrate and top surfaces of each of the plurality of first semiconductor chips, and a plurality of first adhesion layers on bottom surfaces of each of the plurality of first semiconductor chips. The second stack may include a plurality of second semiconductor chips that are stacked on the first stack in ascending stepwise fashion offset in a second direction that is different from the first direction, and a plurality of second adhesion layers on bottom surfaces of the plurality of second semiconductor chips. Each of the first wires may include one end coupled to the top surface of one of the plurality of first semiconductor chips, and a wire loop that extends from the end. A level

difference between the one end of the first wire and a topmost portion of the wire loop may be greater than a thickness of an adjacent one of the plurality of first adhesion layers.

(9) According to some embodiments of the present inventive concepts, a semiconductor package may include a plurality of first memory chips that are stacked on a substrate in ascending stepwise fashion in a first direction, each of the plurality of first memory chips having a first wire that extends between the substrate and a top surface of the first memory chip, and a first adhesion layer on a bottom surface of the first memory chip. A logic chip may be spaced apart in the first direction from the first memory chips, the logic chip having a second wire that extends between the substrate and a top surface of the logic chip. A spacer may be on the logic chip, the spacer having a second adhesion layer on a bottom surface of the spacer. A plurality of second memory chips may be stacked in ascending stepwise fashion offset in a second direction on the plurality of first memory chips and the spacer, the second direction being different from the first direction. A molding layer may be on the substrate, the molding layer covering the plurality of first memory chips, the logic chip, the spacer, and the second memory chips, and a plurality of external terminals on a bottom surface of the substrate. An entirety of the first wire may be spaced apart from the first adhesion layers adjacent to the first wire. At least a portion of the second wire may be inside the second adhesion layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1** and **2** illustrate cross-sectional views of a semiconductor package according to some embodiments of the present inventive concepts.

(2) FIG. **3** illustrates a plan view of a semiconductor package according to some embodiments of the present inventive concepts.

(3) FIGS. **4** and **5** illustrate cross-sectional views of a semiconductor package according to some embodiments of the present inventive concepts.

(4) FIG. **6** illustrates a plan view of a semiconductor package according to some embodiments of the present inventive concepts.

(5) FIGS. **7** and **8** illustrate cross-sectional views of a semiconductor package according to some embodiments of the present inventive concepts.

(6) FIG. **9** illustrates a plan view of a semiconductor package according to some embodiments of the present inventive concepts.

(7) FIG. **10** illustrates a cross-sectional view of a semiconductor package according to some embodiments of the present inventive concepts.

DETAILED DESCRIPTION OF EMBODIMENTS

(8) The following will now describe a semiconductor package according to the present inventive concepts with reference to the accompanying drawings.

(9) FIGS. **1** and **2** illustrate cross-sectional views of a semiconductor package according to some embodiments of the present inventive concepts. FIG. **3** illustrates a plan view of a semiconductor package according to some embodiments of the present inventive concepts. In FIG. **3**, for convenience of description, a second chip stack is illustrated to include only one semiconductor chip that is a lowermost one of third semiconductor chips.

(10) Referring to FIGS. **1** and **3**, a package substrate **100** may be provided. The package substrate **100** may be a printed circuit board (PCB) having a signal pattern provided on a top surface thereof. The signal pattern may include signal pads **110**, **120**, and **130**. The package substrate **100** may have a structure in which at least one dielectric pattern and at least one wiring pattern are alternately stacked. The package substrate **100** may be provided with external terminals **105** on a bottom surface thereof. The external terminals **105** may include a solder ball, a solder bump, or a solder

pad. Based on type of the external terminals **105**, the semiconductor package **10** may have a ball grid array (BGA) type, a fine ball-grid array (FBGA) type, or a land grid array (LGA) type.

(11) A first chip stack CS**1** may be provided on the package substrate **100**. The first chip stack CS**1** may include first semiconductor chips **200** that are stacked in a third direction D**3** on the package substrate **100**. In this description, a first direction D**1** and a second direction D**2** may be defined as directions that intersect each other while being parallel to a top surface of the package substrate **100**, and the third direction D**3** may be a direction perpendicular to the top surface of the package substrate **100**. The first semiconductor chips **200** may be memory chips. For example, the first semiconductor chips **200** may include a dynamic random-access memory (DRAM). The first semiconductor chips **200** may have a first lateral surface **200***a* and a second lateral surface **200***b* that are opposite to each other in the first direction D**1**. For example, the first lateral surfaces **200***a* of the first semiconductor chips **200** may be positioned in a direction opposite to the first direction D**1**, and the second lateral surfaces **200***b* of the first semiconductor chips **200** may be positioned in the first direction D**1**. The first semiconductor chips **200** may be in an offset stack structure. For example, the first semiconductor chips **200** may be stacked obliquely in the first direction D**1**, which may result in an ascending stepwise shape or a cascade shape. For example, each of the first semiconductor chips **200** may protrude in the first direction D**1** from an underlying first semiconductor chip **200**.

(12) As the first semiconductor chips **200** are stepwise stacked, top surfaces of the first semiconductor chips **200** may be partially exposed. For a single first semiconductor chip **200**, a first connection region CR**1** may be defined to indicate the exposed portion of the top surface of the first semiconductor chip **200**. Stated differently, the first connection region CR**1** of a first semiconductor chip **200** may be free from overlap in the third or vertical direction by another of the first semiconductor chips **200**. Based on an offset stacking direction of the first semiconductor chips **200**, the first connection regions CR**1** may be positioned adjacent to the first lateral surfaces **200***a* of the first semiconductor chips **200**. In this description, the expression "offset stacking direction" may be defined to refer to a shift direction relative to an underlying semiconductor chip when semiconductor chips are stacked. For example, in FIG. **1**, the first direction D**1** may be an offset stacking direction of the first semiconductor chips **200**. For convenience of description below, a first sub-semiconductor chip **200-1** may be defined to indicate an underlying first semiconductor chip **200** of arbitrary neighboring first semiconductor chips **200**, and a second sub-semiconductor chip **200-2** may be defined to indicate a next overlying first semiconductor chip **200** of arbitrary neighboring first semiconductor chips **200**. The first connection region CR**1** of the first sub-semiconductor chip **200-1** may be positioned in a direction opposite to the first direction D**1** from the second sub-semiconductor chip **200-2**. The top surfaces of the first semiconductor chips **200** may be active surfaces of the first semiconductor chips **200**. For example, the first semiconductor chips **200** may be provided with first chip pads **210** on the first connection regions CR**1** at the top surfaces of the first semiconductor chips **200**. The first chip pads **210** may be connected to integrated circuits of the first semiconductor chips **200**.

(13) The first semiconductor chips **200** may be wire-bonded to the package substrate **100**. The first semiconductor chips **200** may be connected through first connection wires **215** to the package substrate **100**. For example, the first connection wires **215** may connect the first chip pads **210** of the first semiconductor chips **200** to first signal pads **110** of the package substrate **100**. For example, the first connection wires **215** may each include a first end **215***a* coupled to the first chip pad **210**, a second end **215***b* coupled to the first signal pad **110**, and a wire loop **215***c* that connects the first end **215***a* to the second end **215***b*. When viewed in a plan view, the first signal pads **110** may be positioned in a direction opposite to the first direction D**1** from the first chip stack CS**1**. The first semiconductor chips **200** may be correspondingly provided with first adhesion layers **220** on the bottom surfaces thereof. Each of the first semiconductor chips **200** may be attached thorough the first adhesion layer **220** to an underlying another first semiconductor chip **200**. For example, the

second sub-semiconductor chip **200-2** may be attached to a top surface of the first sub-semiconductor chip **200-1** through the first adhesion layer **220** provided on a bottom surface of the second sub-semiconductor chip **200-2**. In this case, the second sub-semiconductor chip **200-2** may expose the first connection region CR**1** of the first sub-semiconductor chip **200-1**. A lowermost one (referred to hereinafter as a lowermost first semiconductor chip) of the first semiconductor chips **200** may be attached through the first adhesion layer **220** to the package substrate **100**. The first adhesion layer **220** may have a first thickness t**1** of about 5 μm to about 30 μm. For example, the first thickness t**1** of the first adhesion layer **220** may be equal to or less than about 20 μm. The first adhesion layers **220** may include a die attach film (DAF).

(14) The first connection wires **215** may not be inserted into the first adhesion layers **220**. For example, the first chip pad **210** of the first sub-semiconductor chip **200-1** may be spaced apart from the second sub-semiconductor chip **200-2** in a direction opposite to the first direction D**1**. In such cases, the first chip pad **210** of the first sub-semiconductor chip **200-1** may not be covered with the first adhesion layer **220** of the second sub-semiconductor chip **200-2**. The first connection wire **215** of the first sub-semiconductor chip **200-1** may be spaced apart from the second sub-semiconductor chip **200-2** in a direction opposite to the first direction D**1**. The first connection wire **215** of the first sub-semiconductor chip **200-1** may be spaced apart from the first adhesion layer **220** of the second sub-semiconductor chip **200-2**. As the first connection wire **215** is not inserted into the first adhesion layer **220** of the second sub-semiconductor chip **200-2**, the first adhesion layer **220** may have a small thickness regardless of placement, length, and an extending direction of the first connection wire **215**. For example, a level difference LD between the first end **215***a* and a topmost portion of the wire loop **215***c* of the first connection wire **215** coupled to the first chip pad **210** of the first sub-semiconductor chip **200-1** may be greater than the first thickness t**1** of the first adhesion layer **220** of the second sub-semiconductor chip **200-2**. The level difference LD may mean a difference in height in the third direction D**3** from the top surface of the package substrate **100**. A range of about 40 μm to about 100 μm may be given as the level difference LD of the first connection wire **215** coupled to the first sub-semiconductor chip **200-1**. The first thickness t**1** of the first adhesion layer **220** may be equal to or less than about 20 μm.

(15) When the first semiconductor chips **200** of the first chip stack CS**1** are vertically stacked, the first chip pad **210** of the first semiconductor chip **200** may be covered with the first adhesion layer **220** of an overlying first semiconductor chip **200**. In this case, the first connection wires **215** may extend from inside the first adhesion layers **220** toward outside the first chip stack CS**1**. Therefore, the first adhesion layers **220** may become thicker as much as heights of the first connection wires **215**, which may result in an increase in thickness of the first chip stack CS**1**.

(16) According to some embodiments of the present inventive concepts, as the first semiconductor chips **200** of the first chip stack CS**1** are in an offset stack structure, the first connection wires **215** may not be inserted into adjacent ones of the first adhesion layers **220** and thus the first adhesion layers **220** may have a relatively small thickness. Thus, the first connection wires **215** may be spaced apart from the first adhesion layers. Accordingly, the first chip stack CS**1** may decrease in thickness and a semiconductor package may be miniaturized.

(17) Referring still to FIGS. **1** to **3**, a second semiconductor chip **300** may be on the package substrate **100**. The second semiconductor chip **300** may be spaced apart in the first direction D**1** from the first chip stack CS**1**. For example, the second semiconductor chip **300** may face the second lateral surfaces **200***b* of the first semiconductor chips **200**. The second semiconductor chip **300** may have a thickness greater than those of the first semiconductor chips **200**. The thickness of the second semiconductor chip **300** may be less than a total thickness of the first chip stack CS**1**. A width and planar shape of the second semiconductor chip **300** may be less than those of the first semiconductor chips **200**. The present inventive concepts, however, are not limited thereto. The second semiconductor chip **300** may be a logic chip. For example, the second semiconductor chip **300** may include a controller.

(18) The second semiconductor chip **300** may have a top surface as an active surface. For example, the second semiconductor chip **300** may be provided with second chip pads **310** on the top surface thereof. The second chip pads **310** may be connected to an integrated circuit of the second semiconductor chip **300**.

(19) The second semiconductor chip **300** may be wire-bonded to the package substrate **100**. The second semiconductor chip **300** may be connected through second connection wires **315** to the package substrate **100**. For example, the second connection wires **315** may connect the second chip pads **310** of the second semiconductor chip **300** to second signal pads **120** of the package substrate **100**. The second signal pads **120** may be positioned either between the first chip stack CS**1** and the second semiconductor chip **300** or in the first direction D**1** from the second semiconductor chip **300**. Alternatively, differently from that shown, the second signal pads **120** may be positioned away from the second semiconductor chip **300** in the second direction D**2** or in a direction opposite to the second direction D**2**.

(20) A second adhesion layer **320** may be provided on a bottom surface of the second semiconductor chip **300**. The second semiconductor chip **300** may be attached through the second adhesion layer **320** to the package substrate **100**. The second adhesion layer **320** may include a die attach film (DAF).

(21) A spacer **400** may be provided on the second semiconductor chip **300**. The spacer **400** may have a width the same as that of the second semiconductor chip **300**. The second semiconductor chip **300** and the spacer **400** may be vertically aligned with each other. For example, the second semiconductor chip **300** may have lateral surfaces aligned with those of the spacer **400**. According to some embodiments, as shown in FIG. **2**, the width of the spacer **400** may be greater than that of the second semiconductor chip **300**. In this case, when viewed in a plan view, the spacer **400** may overlap an entirety of the second semiconductor chip **300**. Alternatively, the spacer **400** may have a width less than that of the second semiconductor chip **300**. In this case, when viewed in a plan view, the second semiconductor chip **300** may overlap an entirety of the spacer **400**. The spacer **400** may be provided to compensate (or reduce) a step difference between the first chip stack CS**1** and the second semiconductor chip **300**. For example, the spacer **400** may have a top surface located at the same level as that of a top surface of the first chip stack CS**1** or a top surface of an uppermost one (referred to hereinafter as an uppermost first semiconductor chip) of the first semiconductor chips **200**. An interval between the uppermost first semiconductor chip **200** and the spacer **400** may range from about 500 μm to about 5,000 μm. The spacer **400** may include bulk silicon (Si).

(22) The spacer **400** may be attached through a third adhesion layer **420** to the top surface of the second semiconductor chip **300**. The third adhesion layer **420** may cover the entirety of the top surface of the second semiconductor chip **300**. In such cases, the second chip pads **310** of the second semiconductor chip **300** may be covered with the third adhesion layer **420**. The second connection wire **315** may extend from the second chip pad **310** to the second signal pad **120** after passing through a space between the second semiconductor chip **300** and the spacer **400** and over one side of the second semiconductor chip **300**. For example, portions of the second connection wires **315** may be inserted into the third adhesion layer **420**. The third adhesion layers **420** may include a die attach film (DAF).

(23) To secure a space where the second connection wires **315** are provided, a large interval may be provided between the second semiconductor chip **300** and the spacer **400**. For example, the third adhesion layer **420** may have a thickness greater than a level difference between the second chip pad **310** and a topmost portion of the second connection wire **315**. The thickness of the third adhesion layer **420** in which the second connection wire **315** is buried may be greater than that of the first adhesion layer **220** in which the first connection wire **215** is not buried. The thickness of the third adhesion layer **420** may range from about 30 μm to about 80 μm.

(24) In some embodiments, when the spacer **400** is provided to have a width less than that of the second semiconductor chip **300**, the second chip pads **310** may not be covered with the third

adhesion layer **420**. In this case, the second connection wires **315** may not be inserted into the third adhesion layer **420**. Therefore, the third adhesion layer **420** may have a small thickness. The thickness of the third adhesion layer **420** may range from about 5 μm to about 30 μm.
(25) Referring further to FIGS. **1** to **3**, a second chip stack CS**2** may be provided on the first chip stack CS**1** and the spacer **400**.
(26) The second chip stack CS**2** may include third semiconductor chips **500** that are stacked on the first chip stack CS**1** and the spacer **400**. The third semiconductor chips **500** may be memory chips. For example, the third semiconductor chips **500** may include a NAND Flash memory. The third semiconductor chips **500** may be in an offset stack structure. For example, the third semiconductor chips **500** may be stacked obliquely in a direction opposite to the first direction D**1**, which may result in an ascending stepwise shape or a cascade shape. In this case, each of the third semiconductor chips **500** may protrude in a direction opposite to the first direction D**1** from an underlying another third semiconductor chip **500**.
(27) As the third semiconductor chips **500** are stepwise stacked, top surfaces of the third semiconductor chips **500** may be partially exposed. For one third semiconductor chip **500**, a second connection region CR**2** may be defined to indicate the exposed portion of the top surface of the third semiconductor chip **500**. Based on an offset stacking direction of the third semiconductor chips **500**, the second connection regions CR**2** may be positioned adjacent to lateral surfaces positioned in the first direction D**1** of the third semiconductor chips **500**. The top surfaces of the third semiconductor chips **500** may be active surfaces of the third semiconductor chips **500**. For example, the third semiconductor chips **500** may be provided with third chip pads **510** on the second connection regions CR**2** at the top surfaces of the third semiconductor chips **500**. The third chip pads **510** may be connected to integrated circuits of the third semiconductor chips **500**.
(28) The third semiconductor chips **500** may be wire-bonded to the package substrate **100**. The third semiconductor chips **500** may be connected through third connection wires **515** to the package substrate **100**. For example, the third connection wires **515** may connect to each other the third chip pads **510** of neighboring third semiconductor chips **500**. Ones of the third connection wires **515** may connect the third chip pads **510** of a lowermost one of the third semiconductor chips **500** to third signal pads **130** of the package substrate **100**. For convenience of description below, a third sub-semiconductor chip **500-1** may be defined to indicate the lowermost one of the third semiconductor chips **500**. When viewed in a plan view, the third signal pads **130** may be positioned in the first direction D**1** from the second semiconductor chip **300**.
(29) The third semiconductor chips **500** may be correspondingly provided with fourth adhesion layers **520** on bottom surfaces thereof. Each of the third semiconductor chips **500** may be attached thorough the fourth adhesion layer **520** to an underlying another third semiconductor chip **500**. For example, the third semiconductor chips **500** may be attached to each other through the fourth adhesion layers **520**. In such cases, each of the third semiconductor chips **500** may expose the second connection region CR**2** of an underlying another third semiconductor chip **500**. The third sub-semiconductor chip **500-1** positioned at the bottom of the second chip stack CS**2** may be attached through the fourth adhesion layer **520** to the first chip stack CS**1** and the spacer **400**. The fourth adhesion layer **520** may have a second thickness t**2** of about 5 μm to about 30 μm. For example, the second thickness t**2** of the fourth adhesion layer **520** may range from about 5 μm to about 10 μm. The fourth adhesion layers **520** may include a die attach film (DAF).
(30) The second chip stack CS**2** may be supported by the first chip stack CS**1** and the spacer **400**. The following will describe arrangement of the first chip stack CS**1**, the second chip stack CS**2**, the second semiconductor chip **300**, and the spacer **400**.
(31) The first chip stack CS**1** may be below the third sub-semiconductor chip **500-1**. The third sub-semiconductor chip **500-1** may cover a portion of the top surface of the uppermost first semiconductor chip **200** and expose another portion of the top surface of the uppermost first semiconductor chip **200**. For example, the third sub-semiconductor chip **500-1** may cover the top

surface of the uppermost first semiconductor chip **200** and expose the first connection region CR**1** of the uppermost first semiconductor chip **200**. The first chip pad **210** of the uppermost first semiconductor chip **200** may be spaced apart in a direction opposite to the first direction D**1** from the third sub-semiconductor chip **500-1** and the fourth adhesion layer **520** of the third sub-semiconductor chip **500-1**. The first connection wire **215** of the uppermost first semiconductor chip **200** may not be inserted into the fourth adhesion layer **520** of the third sub-semiconductor chip **500-1**. Therefore, the fourth adhesion layer **520** may have a small thickness. As a result, it may be possible to provide a compact-sized semiconductor package. The first connection wire **215** of the uppermost first semiconductor chip **200** may have a topmost portion located at a higher level than that of a top surface of the fourth adhesion layer **520** of the third sub-semiconductor chip **500-1**.

(32) The third sub-semiconductor chip **500-1** may vertically overlap at least a portion of the lowermost first semiconductor chip **200**. As the first semiconductor chips **200** are stacked obliquely in the first direction D**1**, at least a portion of each of the first semiconductor chips **200** may overlap the third sub-semiconductor chip **500-1**. For example, the third sub-semiconductor chip **500-1** may be supported by all of the first semiconductor chips **200**.

(33) The second semiconductor chip **300** and the spacer **400** may be below the third sub-semiconductor chip **500-1**. The third sub-semiconductor chip **500-1** may overlap an entirety of the second semiconductor chip **300** and an entirety of the spacer **400**. The second semiconductor chip **300** and the spacer **400** may be vertically stacked, and the second chip stack CS**2** may be supported by an entire top surface of the spacer **400**, with the result that the second chip stack CS**2** may be strongly supported.

(34) In addition, as the top surface of the spacer **400** has a top surface located at the same level as that of the top surface of the uppermost first semiconductor chip **200**, even when the first semiconductor chips **200** are stacked offset in the first direction D**1** toward the second chip stack CS**2**, there may be less or no likelihood that weight of the second chip stack CS**2** will cause the first chip stack CS**1** to tilt or collapse in the first direction D**1**. Accordingly, it may be possible to provide a semiconductor package with structural rigidity.

(35) A molding layer **600** may be provided on the package substrate **100**. The molding layer **600** may cover the first chip stack CS**1**, the second semiconductor chip **300**, the spacer **400**, and the second chip stack CS**2**. The molding layer **600** may include a dielectric polymeric material, such as epoxy molding compound (EMC).

(36) FIGS. **1** to **3** depict that the first semiconductor chips **200** are stacked obliquely in the first direction D**1**, but the present inventive concepts are not limited thereto.

(37) FIG. **4** illustrates a cross-sectional view of a semiconductor package according to some embodiments of the present inventive concepts. For convenience of description, components the same as those of the embodiments discussed with reference to FIGS. **1** to **3** are allocated the same reference numerals thereto, and a repetitive explanation thereof will be omitted or abridged below. The following description will focus on differences between the embodiments of FIGS. **1** to **3** and other embodiments described below.

(38) Referring to FIG. **4**, the first chip stack CS**1** may be provided on the package substrate **100**. The first chip stack CS**1** may include the first semiconductor chips **200** that are stacked in the third direction D**3** on the package substrate **100**. The first semiconductor chips **200** may be in an offset stack structure. For example, the first semiconductor chips **200** may be stacked obliquely in a direction opposite to the first direction D**1**, which may result in an ascending stepwise shape or a cascade shape. In this case, the second sub-semiconductor chip **200-2** may protrude in a direction opposite to the first direction D**1** from the first sub-semiconductor chip **200-1**.

(39) The stepwise stacking of the first semiconductor chips **200** may expose the first connection regions CR**1** at the top surfaces of the first semiconductor chips **200**. Based on the stacking direction of the first semiconductor chips **200**, the first connection regions CR**1** may be located adjacent to the second lateral surfaces **200***b* positioned in the first direction D**1** of the first

semiconductor chips **200**. The first connection region CR**1** of the first sub-semiconductor chip **200-1** may be positioned in the first direction D**1** from the second sub-semiconductor chip **200-2**. The first chip pads **210** may be provided on the first connection regions CR**1** at the top surfaces of the first semiconductor chips **200**.

(40) The first semiconductor chips **200** may be wire-bonded to the package substrate **100**. The first connection wires **215** may connect the first chip pads **210** of the first semiconductor chips **200** to the first signal pads **110** of the package substrate **100**. The first signal pads **110** may be positioned in the first direction D**1** from the first chip stack CS**1**.

(41) The second chip stack CS**2** may be attached to the first chip stack CS**1** and the spacer **400**. The third sub-semiconductor chip **500-1** positioned at the bottom of the second chip stack CS**2** may be attached through the fourth adhesion layer **520** to the first chip stack CS**1** and the spacer **400**. In this case, based on an offset stacking direction of the first semiconductor chips **200**, the first chip pad **210** of the uppermost first semiconductor chip **200** may be covered with the fourth adhesion layer **520** of the third sub-semiconductor chip **500-1**. The first connection wire **215** of the uppermost first semiconductor chip **200** may extend from the first chip pad **210** toward one of the first signal pads **110** after passing through a space, or the fourth adhesion layer **520**, between the uppermost first semiconductor chip **200** and the third sub-semiconductor chip **500-1** and over one side of the first chip stack CS**1**. For example, the first connection wire **215** of the uppermost first semiconductor chip **200** may be partially inserted into the fourth adhesion layer **520** of the third sub-semiconductor chip **500-1**.

(42) To secure a space where is provided the first connection wire **215** of the uppermost first semiconductor chip **200**, a large interval may be provided between the uppermost first semiconductor chip **200** and the third sub-semiconductor chip **500-1**. The thickness of the fourth adhesion layer **520** in which the first connection wire **215** is buried may be greater than that of the first adhesion layer **220** in which the first connection wire **215** is not buried.

(43) According to some embodiments of the present inventive concepts, the first chip stack CS**1** and the second semiconductor chip **300** may be provided therebetween with the first signal pads **110** for connection with the first chip stack CS**1**, and thus a semiconductor package may have a reduced width in the first direction D**1**. Therefore, a semiconductor package may be miniaturized.

(44) FIG. **5** illustrates a cross-sectional view of a semiconductor package according to some embodiments of the present inventive concepts. FIG. **6** illustrates a plan view of a semiconductor package according to some embodiments of the present inventive concepts.

(45) Referring to FIGS. **5** and **6**, the first chip stack CS**1** may be provided on the package substrate **100**. The first chip stack CS**1** may include the first semiconductor chips **200** that are stacked in the third direction D**3** on the package substrate **100**. The first semiconductor chips **200** may be in an offset stack structure. For example, the first semiconductor chips **200** may be stacked obliquely in the first direction D**1**, which may result in an ascending stepwise shape or a cascade shape. In this case, the second sub-semiconductor chip **200-2** may protrude in the second direction D**2** from the first sub-semiconductor chip **200-1**.

(46) The stepwise stacking of the first semiconductor chips **200** may expose the first connection regions CR**1** at the top surfaces of the first semiconductor chips **200**. Based on an offset stacking direction of the first semiconductor chips **200**, the first connection region CR**1** may be located adjacent to lateral surfaces of the first semiconductor chips **200**, which lateral surfaces are positioned in a direction opposite to the second direction D**2**. The first connection region CR**1** of the first sub-semiconductor chip **200-1** may be positioned from the second sub-semiconductor chip **200-2** in a direction opposite to the second direction D**2**. The first chip pads **210** may be provided on the first connection regions CR**1** at the top surfaces of the first semiconductor chips **200**.

(47) The first semiconductor chips **200** may be wire-bonded to the package substrate **100**. The first connection wires **215** may connect the first chip pads **210** of the first semiconductor chips **200** to the first signal pads **110** of the package substrate **100**. The first signal pads **110** may be positioned

from the first chip stack CS**1** in a direction opposite to the second direction D**2**.
(48) According to some embodiments of the present inventive concepts, as the first chip stack CS**1** is provided to have an offset stacking direction (e.g., the second direction D**2**) that intersects an offset stacking direction (e.g., the first direction D**1**) of the second chip stack CS**2**, a semiconductor package may be free of excessive increase in width in the first direction D**1** or the second direction D**2**. In addition, when viewed in a plan view, an electronic device AED, such as additional semiconductor device or passive device, may be mounted on a region where neither the first chip stack CS**1** nor the second chip stack CS**2** is provided, and thus a semiconductor package may increase in integration.
(49) FIG. **7** illustrates a cross-sectional view of a semiconductor package according to some embodiments of the present inventive concepts.
(50) Referring to FIG. **7**, the first chip stack CS**1** may be provided on the package substrate **100**. The first chip stack CS**1** may include the first semiconductor chips **200** that are stacked in the third direction D**3** on the package substrate **100**. The first semiconductor chips **200** may be in an offset stack structure. For example, the first semiconductor chips **200** may be stacked obliquely in the first direction D**1**, which may result in an ascending stepwise shape. In such cases, a certain first semiconductor chip **200** may protrude in the first direction D**1** from an underlying another first semiconductor chip **200**.
(51) The spacer **400** may be on the package substrate **100**. The spacer **400** may be spaced apart in the first direction D**1** from the first chip stack CS**1**. For example, the spacer **400** may face the second lateral surfaces **200***b* of the first semiconductor chips **200**. The spacer **400** may have a thickness substantially the same as that of the first chip stack CS**1**. For example, the top surface of the spacer **400** may be located at the same level as that of the top surface of the first chip stack CS**1**. The spacer **400** may be attached through the third adhesion layer **420** to the top surface of the package substrate **100**.
(52) The second semiconductor chip **300** may be on the package substrate **100**. The second semiconductor chip **300** may be between the first chip stack CS**1** and the spacer **400**. The thickness of the second semiconductor chip **300** may be greater than the thickness of the first semiconductor chip **200** and less than the total thickness of the first chip stack CS**1**. Based on an offset stacking direction of the first chip stack CS**1**, the second semiconductor chip **300** may be adjacent to the first chip stack CS**1**. For example, the second connection wire **315** of the second semiconductor chip **300** may be provided in a space below the uppermost first semiconductor chip **200**, or a portion of the second semiconductor chip **300** may be in a space below the uppermost first semiconductor chip **200**. According to some embodiments of the present inventive concepts, the second semiconductor chip **300** may be positioned close to the first chip stack CS**1**, and there may be a reduction in area occupied by the first chip stack CS**1** and the second semiconductor chip **300**. Therefore, a semiconductor package may be miniaturized.
(53) The second chip stack CS**2** may be attached to the first chip stack CS**1** and the spacer **400**. The third sub-semiconductor chip **500-1** positioned at the bottom of the second chip stack CS**2** may be attached through the fourth adhesion layer **520** to the first chip stack CS**1** and the spacer **400**.
(54) According to some embodiments of the present inventive concepts, the second chip stack CS**2** may be supported by the spacer **400** and the first chip stack CS**1**. For example, on the package substrate **100**, the spacer **400** may directly support the second chip stack CS**2**. Therefore, the second chip stack CS**2** may be strongly supported. Thus, a semiconductor package may be provided to have improved structural stability.
(55) FIG. **8** illustrates a cross-sectional view of a semiconductor package according to some embodiments of the present inventive concepts. FIG. **9** illustrates a plan view of a semiconductor package according to some embodiments of the present inventive concepts. In FIG. **9**, for convenience of description, a second chip stack is illustrated to include only one semiconductor chip that is a lowermost one of third semiconductor chips.

(56) Referring to FIGS. **8** and **9**, the first chip stack CS**1** may be provided on the package substrate **100**. The first chip stack CS**1** may include the first semiconductor chips **200** that are stacked in the third direction D**3** on the package substrate **100**. The first semiconductor chips **200** may be in an offset stack structure. For example, the first semiconductor chips **200** may be stacked obliquely in the first direction D**1**, which may result in an ascending stepwise shape. In such cases, a certain first semiconductor chip **200** may protrude in the first direction D**1** from an underlying another first semiconductor chip **200**.
(57) A third chip stack CS**3** may be provided on the package substrate **100**. The third chip stack CS**3** may be positioned in the first direction D**1** from the first chip stack CS**1**. The third chip stack CS**3** may include fourth semiconductor chips **700** that are stacked in the third direction D**3** on the package substrate **100**. The fourth semiconductor chips **700** may be memory chips. For example, the fourth semiconductor chips **700** may include a dynamic random-access memory (DRAM).
(58) The fourth semiconductor chips **700** may be in an offset stack structure. For example, the fourth semiconductor chips **700** may be stacked obliquely in a direction opposite to the first direction D**1**, which may result in an ascending stepwise shape or a cascade shape. In such cases, each of the fourth semiconductor chips **700** may protrude in a direction opposite to the first direction D**1** from an underlying another fourth semiconductor chip **700**. Such configuration may cause that an interval between the first semiconductor chips **200** and the fourth semiconductor chips **700** may decrease with increasing distance from the package substrate **100**. The present inventive concepts, however, are not limited thereto, and an offset stacking direction of the fourth semiconductor chips **700** may not be opposite to that of the first semiconductor chips **200**. For example, the fourth semiconductor chips **700** may be stacked obliquely in the first direction D**1** or in the second direction D**2**. The third chip stack CS**3** may have a top surface located at the same level as that of the top surface of the first chip stack CS**1**.
(59) The stepwise stacking of the fourth semiconductor chips **700** may expose partial regions at top surfaces of the fourth semiconductor chips **700**. Based on an offset stacking direction of the fourth semiconductor chips **700**, the exposed partial regions may be adjacent to lateral surfaces positioned in the first direction D**1** of the fourth semiconductor chips **700**. The fourth semiconductor chips **700** may be provided with fourth chip pads **710** on the exposed partial regions.
(60) The fourth semiconductor chips **700** may be wire-bonded to the package substrate **100**. A plurality of fourth connection wires **715** may connect the fourth chip pads **710** of the fourth semiconductor chips **700** to fourth signal pads **140** of the package substrate **100**. The fourth signal pads **140** may be positioned in the first direction D**1** from the third chip stack CS**3**.
(61) The fourth semiconductor chips **700** may be provided with fifth adhesion layers **720** on bottom surfaces thereof. Each of the fourth semiconductor chips **700** may be attached through the fifth adhesion layer **720** to an underlying another fourth semiconductor chip **700**. A lowermost one of the fourth semiconductor chips **700** may be attached through the fifth adhesion layer **720** to the package substrate **100**. The fifth adhesion layers **720** may include a die attach film (DAF).
(62) The second semiconductor chip **300** may be on the package substrate **100**. The second semiconductor chip **300** may be between the first chip stack CS**1** and the third chip stack CS**3**. The thickness of the second semiconductor chip **300** may be greater than the thickness of the first semiconductor chip **200** and less than the total thickness of the first chip stack CS**1**. Based on an offset stacking direction of the first chip stack CS**1** and of the third chip stack CS**3**, the second semiconductor chip **300** may be adjacent to the first chip stack CS**1** and the third chip stack CS**3**. For example, the second connection wires **315** of the second semiconductor chip **300** may be provided in a space below the uppermost first semiconductor chip **200** or an uppermost fourth semiconductor chip **700**, or a portion of the second semiconductor chip **300** may be in a space below the uppermost first semiconductor chip **200** or the uppermost fourth semiconductor chip **700**. The second semiconductor chip **300** may be attached through the second adhesion layer **320** to the package substrate **100**.

(63) According to some embodiments of the present inventive concepts, a plurality of chip stacks CS**1**, CS**2**, and CS**3** may be provided to increase integration of a semiconductor package, and in addition an area occupied by the first chip stack CS**1**, the second chip stack CS**2**, and the third chip stack CS**3** may be small to reduce a size of the semiconductor package.
(64) The spacer **400** may be provided on the second semiconductor chip **300**. The spacer **400** may have a width equal to or less than that of the second semiconductor chip **300**. The spacer **400** may be provided to compensate (or reduce) a step difference between the second semiconductor chip **300** and each of the first and third chip stacks CS**1** and CS**3**. For example, the spacer **400** may have a top surface located at the same level as that of a top surface of the first chip stack CS**1** (or a top surface of the uppermost first semiconductor chip **200**) and that of a top surface of the third chip stack CS**3** (or a top surface of the uppermost fourth semiconductor chip **700**). The spacer **400** may be attached through the third adhesion layer **420** to the top surface of the second semiconductor chip **300**.
(65) The second chip stack CS**2** may be provided on the first chip stack CS**1**, the third chip stack CS**3**, and the spacer **400**. The second chip stack CS**2** may include the third semiconductor chips **500** that are stacked in the third direction D**3** on the first chip stack CS**1**, the third chip stack CS**3**, and the spacer **400**. The third semiconductor chips **500** may be in an offset stack structure. For example, the third semiconductor chips **500** may be stacked obliquely in the second direction D**2** or in a direction opposite to the second direction D**2**, which may result in an ascending stepwise shape. According to some embodiments, the third semiconductor chips **500** may be stacked vertically aligned with each other.
(66) The stepwise stacking of the third semiconductor chips **500** may expose partial regions at top surfaces of the third semiconductor chips **500**. The third chip pads **510** may be provided on the exposed partial regions of the third semiconductor chips **500**.
(67) The third semiconductor chips **500** may be wire-bonded to the package substrate **100**. The third connection wires **515** may connect the third chip pads **510** of the third semiconductor chips **500** to the third signal pads **130** of the package substrate **100**. The third signal pads **130** may be positioned from the second chip stack CS**2** in the second direction D**2** or in a direction opposite to the second direction D**2**.
(68) The second chip stack CS**2** may be attached to the first chip stack CS**1**, the third chip stack CS**3**, and the spacer **400**. The third sub-semiconductor chip **500-1** positioned at the bottom of the second chip stack CS**2** may be attached through the fourth adhesion layer **520** to the first chip stack CS**1**, the third chip stack CS**3**, and the spacer **400**. In this case, based on a stacking direction of the fourth semiconductor chips **700**, the fourth chip pad **710** of the uppermost fourth semiconductor chip **700** may not be covered with the fourth adhesion layer **520** of the third sub-semiconductor chip **500-1**. The fourth connection wire **715** of the uppermost fourth semiconductor chip **700** may not be inserted into the fourth adhesion layer **520** of the third sub-semiconductor chip **500-1**. Therefore, the fourth adhesion layer **520** may have a small thickness. Therefore, a semiconductor package may be miniaturized.
(69) The third sub-semiconductor chip **500-1** may vertically overlap at least a portion of the lowermost first semiconductor chip **200** and at least a portion of the lowermost fourth semiconductor chip **700**. For example, the second chip stack CS**2** may be supported by all the first semiconductor chips **200** and all the fourth semiconductor chips **700**.
(70) The third sub-semiconductor chip **500-1** may be supported by the spacer **400** positioned between the first chip stack CS**1** and the third chip stack CS**3**. Therefore, even when the first semiconductor chips **200** and the fourth semiconductor chips **700** are stacked offset toward each other, there may be less or no likelihood that weight of the second chip stack CS**2** will cause the first and third chip stacks CS**1** and CS**3** to collapse toward each other. Accordingly, it may be possible to provide a semiconductor package with structural rigidity.
(71) FIG. **10** illustrates a cross-sectional view of a semiconductor package according to some

embodiments of the present inventive concepts.
(72) Referring to FIG. **10**, a fourth chip stack CS**4** may be provided between the package substrate **100** and the first chip stack CS**1**. The fourth chip stack CS**4** may include fifth semiconductor chips **800** that are stacked in the third direction D**3** on the package substrate **100**. The fifth semiconductor chips **800** may be in an offset stack structure. For example, an offset stacking direction of the fifth semiconductor chips **800** may be opposite to that of the first semiconductor chips **200**. For example, the fifth semiconductor chips **800** may be stacked obliquely in a direction opposite to the first direction D**1**, which may result in an ascending stepwise shape. In such cases, a certain fifth semiconductor chip **800** may protrude in a direction opposite to the first direction D**1** from an underlying another fifth semiconductor chip **800**.
(73) The stepwise stacking of the fifth semiconductor chips **800** may expose partial regions at top surfaces of the fifth semiconductor chips **800**. Based on an offset stacking direction of the fifth semiconductor chips **800**, the exposed partial regions may be adjacent to lateral surfaces positioned in the first direction D**1** of the fifth semiconductor chips **800**. The fifth semiconductor chips **800** may be provided with fifth chip pads **810** on the exposed partial regions.
(74) The fifth semiconductor chips **800** may be wire-bonded to the package substrate **100**. A plurality of fifth connection wires **815** may connect the fifth chip pads **810** of the fifth semiconductor chips **800** to fifth signal pads **150** of the package substrate **100**. The fifth signal pads **150** may be positioned in the first direction D**1** from the fourth chip stack CS**4**.
(75) The fifth semiconductor chips **800** may be provided with sixth adhesion layers **820** on bottom surfaces thereof. Each of the fifth semiconductor chips **800** may be attached through the sixth adhesion layer **820** to an underlying another fifth semiconductor chip **800**. A lowermost one of the fifth semiconductor chips **800** may be attached through the sixth adhesion layer **820** to the package substrate **100**. The sixth adhesion layers **820** may include a die attach film (DAF).
(76) The first chip stack CS**1** may be on an uppermost fifth semiconductor chip **800**. The first chip stack CS**1** may include the first semiconductor chips **200** that are stacked in the third direction D**3** on the fourth chip stack CS**4**. The first semiconductor chips **200** may be in an offset stack structure. For example, the first semiconductor chips **200** may be stacked obliquely in the first direction D**1**, which may result in an ascending stepwise shape. In such cases, the lowermost first semiconductor chip **200** may overlap at least a portion of the lowermost fifth semiconductor chip **800**. Thus, the first chip stack CS**1** may be supported by all the fifth semiconductor chips **800**.
(77) The second chip stack CS**2** may be attached to the first chip stack CS**1** and the spacer **400**. The third sub-semiconductor chip **500-1** positioned at the bottom of the second chip stack CS**2** may be attached through the fourth adhesion layer **520** to the first chip stack CS**1** and the spacer **400**.
(78) The third sub-semiconductor chip **500-1** may vertically overlap at least a portion of the lowermost first semiconductor chip **200** and at least a portion of the lowermost fifth semiconductor chip **800**. For example, the second chip stack CS**2** may be supported by all the first semiconductor chips **200** and all the fifth semiconductor chips **800**, and accordingly it may be possible to provide a semiconductor package with structural rigidity.
(79) For a semiconductor package according to some embodiments of the present inventive concepts, as semiconductor chips of a lower chip stack are in an offset stack structure, connection wires of the lower chip stack may not be inserted into adhesion layers, and thus the adhesion layers may have their small thicknesses. Accordingly, the lower chip stack may decrease in thickness and the semiconductor package may be miniaturized.
(80) In addition, as a spacer has a top surface located at the same level as that of a top surface of the lower chip stack, even when semiconductor chips of the lower chip stack are stacked offset in one direction, there may be less or no likelihood that weight of an upper chip stack will cause the lower chip stack to collapse in the one direction. Thus, the semiconductor package may be provided to have improved structural stability.
(81) Although the present inventive concepts have been described in connection with the

embodiments of the present inventive concepts illustrated in the accompanying drawings, it will be understood by one of ordinary skill in the art that variations in form and detail may be made therein without departing from the spirit and essential feature of the present inventive concepts. The above disclosed embodiments should thus be considered illustrative and not restrictive.

Claims

1. A semiconductor package, comprising: a first chip stack that includes a plurality of first semiconductor chips on a substrate, the plurality of first semiconductor chips is in an offset stack structure and stacked such that a connection region is exposed at a top surface of each of the plurality of first semiconductor chips; a second semiconductor chip on the substrate and horizontally spaced apart from the first chip stack; a spacer on the second semiconductor chip; and a second chip stack that includes a plurality of third semiconductor chips on the first chip stack and the spacer, the plurality of third semiconductor chips is in an offset stack structure, wherein each of the plurality of first semiconductor chips includes: a first chip pad on the connection region; and a first wire that extends between the first chip pad and the substrate, wherein the first wire of an uppermost one of the plurality of first semiconductor chips is horizontally spaced apart from a lowermost one of the plurality of third semiconductor chips wherein the plurality of first semiconductor chips that are vertically adjacent to each other are attached to each other through a plurality of first adhesion layers provided on bottom surfaces of the plurality of first semiconductor chips wherein: the spacer is attached through a second adhesion layer to a top surface of the second semiconductor chip, the second semiconductor chip has a second wire that extends between the second semiconductor chip and the substrate from the top surface of the second semiconductor chip, a portion of the second wire being positioned in the second adhesion layer under the spacer, and a thickness of the second adhesion layer is greater than a thickness of one of the plurality of first adhesion layers.
2. The semiconductor package of claim 1, wherein the first wire of each of the plurality of first semiconductor chips is not inserted into the plurality of first adhesion layers.
3. The semiconductor package of claim 1, wherein the first wire of each of the plurality of first semiconductor chips includes: a first end coupled to the first chip pad; a second end coupled to the substrate; and a wire loop that connects the first end and the second end to each other, wherein a level difference between the first end and a topmost portion of the wire loop is greater than a thickness of one of the plurality of first adhesion layers.
4. The semiconductor package of claim 1, wherein the thickness of one of the plurality of first adhesion layers is in a range of about 5 μm to about 20 μm, and the thickness of the second adhesion layer is in a range of about 30 μm to about 80 μm.
5. The semiconductor package of claim 1, wherein the second chip stack is attached through a third adhesion layer to a top surface of the first chip stack and to a top surface of the spacer.
6. The semiconductor package of claim 5, wherein a topmost portion of the first wire of the uppermost one of the plurality of first semiconductor chips is at a level higher than a level of a top surface of the third adhesion layer.
7. The semiconductor package of claim 1, wherein the plurality of first semiconductor chips are stacked on the substrate in ascending stepwise fashion offset in a first direction toward the second semiconductor chip.
8. The semiconductor package of claim 7, wherein the connection region is adjacent to one lateral surface in a second direction of one of the plurality of first semiconductor chips, the second direction being opposite to the first direction, and the first wire extends from the connection region of one of the plurality of first semiconductor chips toward one side in the second direction of the plurality of first semiconductor chips and has a coupling to a substrate pad of the substrate.
9. The semiconductor package of claim 1, wherein the plurality of first semiconductor chips are

stacked on the substrate in ascending stepwise fashion in a third direction away from the second semiconductor chip.

10. The semiconductor package of claim 9, wherein the connection region is adjacent to one lateral surface in a fourth direction of the plurality of first semiconductor chips, the fourth direction being opposite to the third direction, and the first wire extends between the connection region of one of the plurality of first semiconductor chips and a space between the first chip stack and the second semiconductor chip and has coupling to a substrate pad of the substrate.

11. The semiconductor package of claim 1, wherein the lowermost one of the plurality of third semiconductor chips vertically overlaps an entirety of the second semiconductor chip and overlaps at least a portion of a lowermost one of the plurality of first semiconductor chips.

12. A semiconductor package, comprising: a substrate and a plurality of spaced apart signal pads on the substrate; a first stack on the substrate; a second stack on the first stack; and a spacer on one side of the first stack and between the substrate and the second stack, wherein the first stack includes: a plurality of first semiconductor chips that are stacked on the substrate in ascending stepwise fashion offset in a first direction; a plurality of first wires that extend between the substrate and top surfaces of each of the plurality of first semiconductor chips; and a plurality of first adhesion layers on bottom surfaces of each of the plurality of first semiconductor chips, wherein the second stack includes: a plurality of second semiconductor chips that are stacked on the first stack in ascending stepwise fashion offset in a second direction, the second direction being different from the first direction; and a plurality of second adhesion layers on bottom surfaces of the second semiconductor chips, wherein each of the first wires includes: a first end coupled to the top surface of a respective one of the plurality of first semiconductor chips; a wire loop that extends from the first end; and a second end coupled to a respective one of the plurality of spaced apart signal pads on the substrate, wherein a level difference between the first end of the first wire and a topmost portion of the wire loop is greater than a thickness of an adjacent one of the plurality of first adhesion layers.

13. The semiconductor package of claim 12, wherein the second direction is opposite the first direction, and the first wire is coupled to one of the plurality of first semiconductor chips and is spaced apart in the second direction from another of the plurality of first semiconductor chips, the another of the plurality of first semiconductor chips being attached to the one of the plurality of first semiconductor chips.

14. The semiconductor package of claim 12, wherein a top surface of an uppermost one of the plurality of first semiconductor chips is at a level the same as a level of a top surface of the spacer.

15. The semiconductor package of claim 12, further comprising a third semiconductor chip between the substrate and the spacer, wherein the spacer is attached to the third semiconductor chip through a third adhesion layer provided on a bottom surface of the spacer, and wherein the third semiconductor chip has a second wire that extends between the substrate and a top surface of the third semiconductor chip, a portion of the second wire extending into the third adhesion layer.

16. The semiconductor package of claim 15, wherein a thickness of the third adhesion layer is greater than the thickness of ones of the plurality of first adhesion layers and greater than a thickness of ones of the plurality of second adhesion layers, and the level difference between the first end of the first wire and a topmost portion of the wire loop is greater than a thickness of ones of the plurality of second adhesion layers.

17. A semiconductor package, comprising: a plurality of first memory chips that are stacked on a substrate in ascending stepwise fashion offset in a first direction, each of the plurality of first memory chips having a first wire that extends between the substrate and a top surface of the first memory chip, and a first adhesion layer on a bottom surface of the first memory chip; a logic chip that is spaced apart in the first direction from the plurality of first memory chips, the logic chip having a second wire that extends between the substrate and a top surface of the logic chip; a spacer on the logic chip, the spacer having a second adhesion layer on a bottom surface of the

spacer; a plurality of second memory chips that are stacked in ascending stepwise fashion offset in a second direction on the plurality of first memory chips and the spacer, the second direction being different from the first direction; a molding layer on the substrate, the molding layer covering the plurality of first memory chips, the logic chip, the spacer, and the second memory chips; and a plurality of external terminals on a bottom surface of the substrate, wherein an entirety of the first wire is spaced apart from the first adhesion layers adjacent to the first wire, and wherein at least a portion of the second wire is inside the second adhesion layer and under the spacer.
18. The semiconductor package of claim 17, wherein a thickness of the first adhesion layer is less than a thickness of the second adhesion layer.